Jose Carvajal-Beltran

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Professional Summary

Materials Science and Engineering undergraduate at MIT with a physics concentration and active research experience in volumetric additive manufacturing, carbon nanostructure composites, and laser safety metrology. Comfortable contributing to lab environments that span computation, fabrication, and experimental characterization.

Education

Massachusetts Institute of Technology (MIT) Cambridge, MA
B.S. Materials Science and Engineering (Course 3-A), Physics Concentration. *May 2029*
Relevant Coursework: 3.091 Solid-State Chemistry, 18.03 Differential Equations, 8.012 Physics I

Experience

Undergraduate Researcher, CNT Growth for BNNT Development Sep. 2026 – Present
MIT UROP, necstlab
- Grow CNT forests via chemical vapor deposition and plasma-etch the arrays to support development of parametric BNNT arrays.
- Measure CNT forest height and mass, maintain structured sample records, and support surface-area and functional-group characterization.

Undergraduate Teaching Assistant & Grader, Physics 8.01 Sep. 2026 – Dec. 2026
MIT Department of Physics
- Support the course's highest-enrollment section in MIT's TEAL format, facilitating collaborative problem-solving and group activities in introductory classical mechanics.
- Provide individualized guidance during class and grade weekly problem sets for accuracy, reasoning, and clear presentation.

Research Intern, Carbon Nanostructure Polymer Composites Jun. 2026 – Aug. 2026
Universidad Anáhuac Mayab, MIT MISTI Program
- Selected through MIT MISTI for a placement under Dr. Andrés I. Oliva-Avilés, contributing to a SECIHTI-funded national research initiative.
- Fabricated carbon nanotube and graphene polymer nanocomposites across a weight percent series, tuning ultrasonication parameters and applying electric-field-assisted casting to align filler networks during cure.
- Characterized piezoresistive response across multiple variables and presented findings via literature review.

Undergraduate Researcher, Volumetric Additive Manufacturing Feb. 2026 – May 2026
MIT UROP, Wallin Group
- Optimized azimuthal and polar rotation angles for VAM via spatial frequency analysis on benchmark CAD geometries.
- Extracted orientation heuristics across seven geometries and resolved an inscribed-sphere fitting artifact, reducing print preparation iterations upon codebase integration.

Student Shop Mentor Sep. 2025 – May 2026
MIT Edgerton Center
- Trained peers on shop equipment, managed workspace operations and after-hours access, and advised on fabrication projects.

Projects

Consumer Laser Product Compliance May 2026 – Present
FDA Citizen Petition, Docket FDA-2026-P-7822
- Researched, drafted, and filed a Citizen Petition with the FDA under 21 CFR 10.30 seeking enforcement of accessible emission limits against mislabeled consumer laser pointers.

Mu-Zero Air Hockey Table Jul. 2025 – Sep. 2025
Electronics Lead, MIT Edgerton Center
- Wired a 120V AC OEM blower with safety interlocks and built two beam-break sensor circuits on protoboard for automatic scorekeeping; presented to an audience of 80+, including industry guests.

Autorotation and Reentry Modelling Aug. 2021 – May 2024
Independent Researcher
- Modelled flight dynamics of an autorotating maple samara and extended the principles to a theoretical rotor-equipped reentry capsule; presented twice at ISEF, earning two special awards (CIA + Scholarship).

Skills

Computation: Python (NumPy, SciPy, Matplotlib), Simulation and optimization pipeline design, Git, LaTeX

Laboratory and Fabrication: Ultrasonication and nanofiller dispersion, Electric-field-assisted casting, Piezoresistive and viscoelastic characterization, Optical power measurement, Analog and digital circuits, Arduino, 3D printing